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Undergraduate Students Individual Project Report

Study on the wave parameter reconstruction based on machine-learning

College: Southampton Ocean Engineering Joint Institute

Program Title: XXX Engineering

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基于机器学习的波浪参数重构研究

学 院 名 称：南安普顿海洋工程联合学院

专 业 名 称：船舶与海洋工程

学 生 姓 名：张三

指 导 教 师：孟巍副教授/Blair Thornton 教授

哈尔滨工程大学

2026 年 5 月

Study on the wave parameter reconstruction based on machine-learning **San Zhang Harbin Engineering University**

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Study on the wave parameter reconstruction based on machine-learning

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基于机器学习的波浪参数重构研究

Preparation and study of XXXXX（**IP Title**）

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哈尔滨工程大学

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ABSTRACT

The study of wave propagation characteristics near the ocean bottom is a prominent field in oceanography.

The thesis XXX.

Keywords: Water wave; Ground wave; ship collision

摘　要

海洋声传播研究长期以来一直是海洋声学学科中的热门领域，尤其是在海底附近的水波和地波传播特性方面，如时域波形和传播损失等。尽管目前在此领域取得了一定进展，但对于水波和地波的分离研究甚少。这些问题的解决不仅对海洋声场分析有重要意义，还对海洋资源开发和海底地震灾害预测具有实用价值。

在这一背景下，本文探讨了XXXX。

结果表明，XXXX。

关键词：水波；底波；模态频散；谱元法；水波分离

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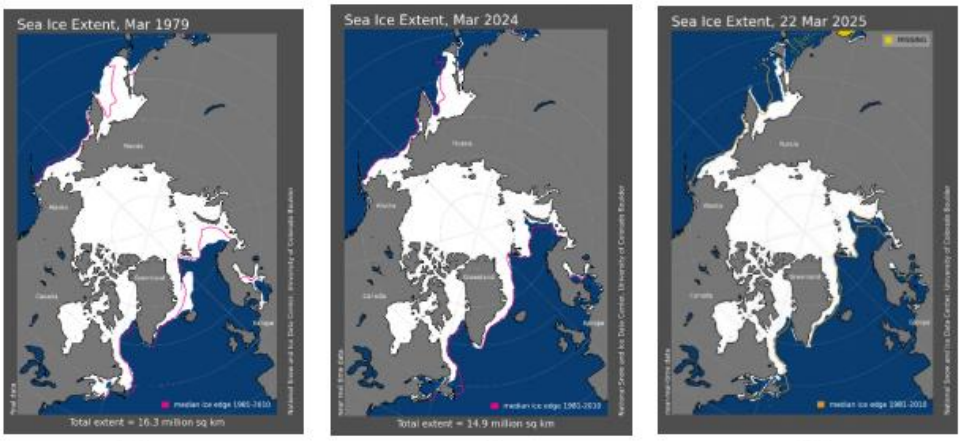
I Introduction

1.1 Background and significance

Oceans cover more than seventy percent of the Earth's surface, serving as the cradle of life, nurturing countless species and abundant energy resources.

（图片和表格注释参考南安普顿书写要求，**查找路径：钉钉全院文件夹——22级毕设——IP 辅导PPT资料——2022级IP 辅导资料**）

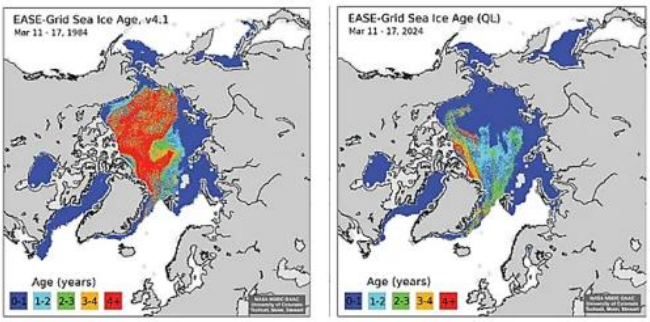


(a) (b) (c)

Fig. 1.1 Arctic sea ice extent for (a) March ,1979 and (b) March ,2024 (c) March 22, 2025 , was 14.33 million square kilometers. The red line in (a) (b) and orange line in (c) show the 1981 to 2010 average extent for that day. (National Snow and Ice Data Center [NSIDC], Accessed 2024)

1.2 Ground waves and water waves

In shallow water environments, the propagation of sound waves is determined by the unique physical properties of the environment, exhibiting distinct propagation characteristics (Pekeris 17).



(a) (b)

Fig. 1.2 Sea ice age for the week of March 11 to March 17 for (a) 1984 and (b) 2024. Data and images from NSIDC EASE-Grid Sea Ice Age, Version 4 (Tschudi et al., 2019a) and Quicklook Arctic Weekly EASE-Grid Sea Ice Age, Version 1 (Tschudi et al., 2019b).

1.3 Project aim and objectives

1.3.1 Project aim

The project focuses on the acoustic propagation characteristics of broadband sound sources in shallow water, aiming to study theoretically and experimentally the method of separating the ground wave from water wave and analyse the transmission efficiency.

1.3.2 Project objectives

To comprehensively understand and simulate the interaction between ground and water waves, the research has established the following specific objectives:

1.4 Research contents

This thesis analyses water and ground waves near the ocean bottom from numerical simulation and experimental data processing perspectives.

II Literature Review

2.1 Review of water wave and ground wave

As early as 1948, Pekeris proposed a method for solving the normal mode acoustic field in the context of exploring shallow water waveguide problems composed of a uniform fluid layer and a semi-infinite liquid ocean bottom layer (Pekeris 43).

2.2 Review of Spectral Element Method

The spectral element method, as a type of generalised finite element method, combines the flexible handling capability of the finite element method with the high-speed convergence characteristics of the pseudo-spectral method, providing an efficient solution for the numerical simulation of sound wave propagation (TANG 65–75).

III Brief Theory and Model

In studying sound propagation in shallow waters, broadband point sources are taken into consideration.

3.1 Normal mode theory

The normal mode theory was first developed by Pekeris as a simple two-layer fluid ocean model (Jensen et al. 102–33).

3.2 Ground/water waves and Airy waves theory

The Pekeris waveguide, proposed by Pekeris, is a waveguide environment consisting of a uniform seawater layer and a liquid semi-infinite space beneath it (Pekeris 1–116)

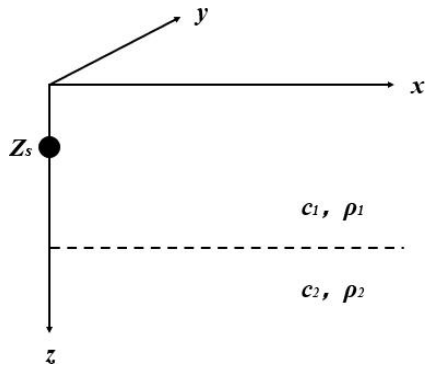


Fig. 3.2 Pekeris waveguide environment

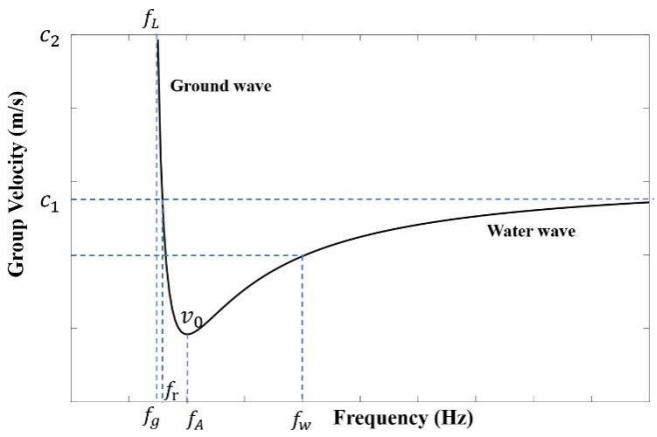


Fig. 3.3 First-Order Mode Group Velocity-Frequency Relationship

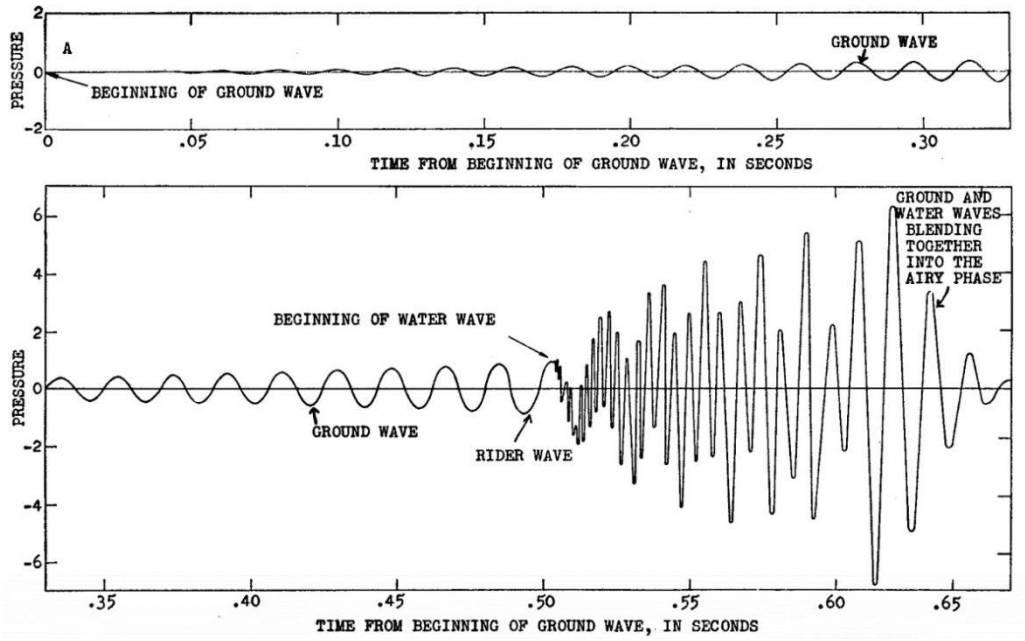


Fig. 3.4 Theoretical contribution of the first mode to the disturbance, from Chaim Leib Pekeris, "Theory of Propagation of Explosive Sound in Shallow Water" (1948 94).

3.3 Spectral Element Method Model based on SPECFEM2D

The core idea of the Spectral Element Method is to first divide the entire computational area into a finite number of elements and then apply the Pseudospectral Method, distributing non-uniform nodes within each element.

3.3.1 SPECFEM2D program configuration

SPECFEM2D is a Spectral Element Method program designed for computations on Linux systems.

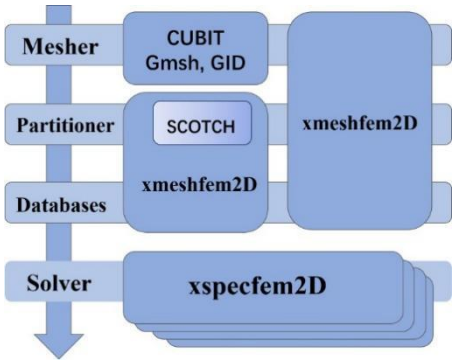


Fig. 3.5 SPECFEM2D Simulation Flowchart

3.3.2 Simulation environment establishment

When conducting ocean acoustic field modelling, it is necessary to determine parameters such as sound speed and density.

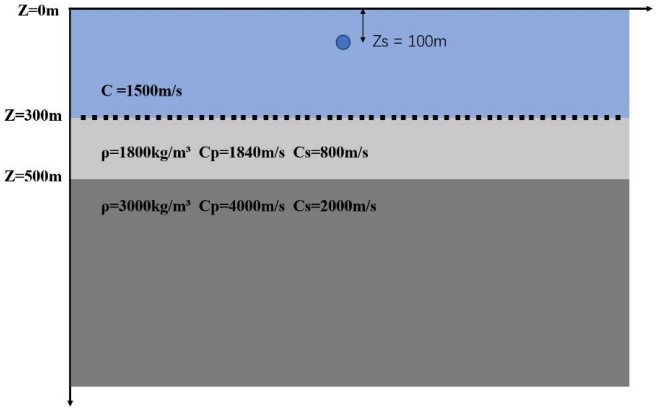


Fig. 3.6 Shallow water wave guide model

3.4 Chapter Summary

This chapter primarily conducts theoretical analyses on the following three aspects:

IV Numerical study of Ground/ Water waves

In order to measure the transmission efficiency of the water wave-ground wave channel, the study utilises the water layer, elastic sedimentary layer, and semi-infinite ocean bottom layer model created in Section 3.3 using the spectral element method.

4.1 Numerical methods and parameter settings

The separation of water waves from ground waves first necessitates the modal resolution of water and ground wave signals, achieved through domain transformation of signals received by the horizontal array stations.

Medium	Air	Snow	Sea ice	Sea water
Absorption coefficient	0.1	2.0	2.5	5.0
Scattering coefficient	2.9	250.0	15.0	10.0
Asymmetric parameter	0.99	0.87	0.92	0.9
Refractive index	1.00	1.49	1.309	1.339

Table 4.1 The inherent optical properties of various media(air, snow, sea ice, and sea water)

4.2 Separation of ground and water waves

4.2.1 Time-domain characteristics analysis

4.2.2 τ-p transformation analysis

The method of t - p transformation involves stacking signals in the xt- domain according to different intercepts t and slopes p (Zhou and Greenhalgh 1133–49).

4.3 Transmission Loss Analysis

As shown in Fig. 4.12, sound waves exhibit significant attenuation characteristics near the source area.

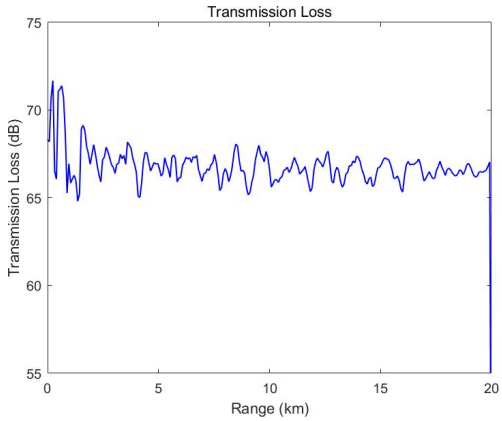


Fig. 4.12 Transmission Loss

4.4 Chapter summary

This chapter aims to improve the comprehension of how sound waves propagate in the horizontally stratified waveguides by utilising the SPECFEM2D for sound field modelling.

V Shallow water experiment analysis

As shown in Chapter 4, it presents a simulation of a three-layer horizontally stratified ocean bottom using the two-dimensional Spectral Element Method.

5.1 Experiment Overview

The College of Underwater Acoustic Engineering at Harbin Engineering University conducted a comprehensive marine experiment in a specific sea area.

5.2 Data Processing and Analysis

Similar to numerical studies, the processing of experimental data primarily relies on the water-ground modal separation technique based on the time-frequency characteristics of signals from horizontal arrays at the ocean bottom interface.

5.3 Chapter Summary

This chapter focuses on the processing and analysis of experimental data, with the aim of separating the ground waves and water waves.

VI Conclusion

The aim of the study is to separate the ground waves and water waves in a horizontally stratified environment.

Reference

参考文献写作规范参考哈佛大学写作规范（查找路径：钉钉全院文件夹——22级毕设——论文模板）

致谢

Acknowledgement

Published papers and research achievements during the pursuit of bachelor's degree

Appendix